Monotonicity of derivability

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: standalone; consumed by the deduction-theorem family (this batch).

Editorial note. The label prop:monotonicity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Proposition 0.1 (Monotonicity)**.** *If* $\Gamma \subseteq \Delta$ *and* $\Gamma \vdash A$*, then* $\Delta \vdash A$*.*

Proof. Any derivation of A from Γ is also derivation of A from Δ. □